

ELE214 - Electronics Laboratory I Preliminary Work 6

1 Questions

1. Determine the range in which the MOSFET is employed as an amplifier. Comment on whether the MOSFET is ON or OFF in this range.
2. Calculate the appropriate R_{P1} (constant) value for the V_{out} voltage to be 7.5 V and explain it in detail. Also, find the voltage gain of the circuit. Take $V_{in} = 50 \sin(\omega t)$ mV, at 100 Hz.

Computer Assignment: Simulate 1-2.

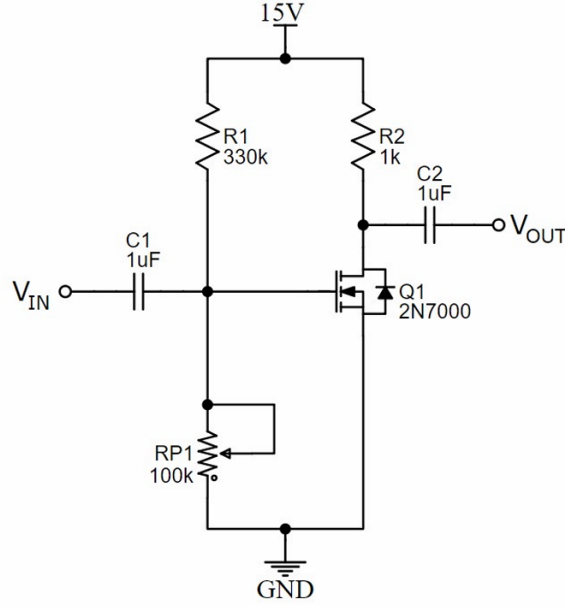


Figure 1: Circuit diagram

Experimental Work

1. Set up the circuit in Figure 1. Adjust the potentiometer to obtain the output voltage of 4 V, 7.5 V, and 11 V (measure the V_{out} using a multimeter). Repeat steps 2 and 3 for $V_{out} = 4$ V, $V_{out} = 7.5$ V, and $V_{out} = 11$ V.
2. Measure the V_{GS} value of the MOSFET.
3. Determine the voltage gain. Take $V_{in} = 50 \sin(\omega t)$ mV at 100 Hz.

2 Answers

1. For the MOSFET to be used as an amplifier, it should be in the saturation mode. In this mode, the MOSFET is ON. Impose the condition for saturation:

$$V_{DS} \geq V_{GS} - V_{TH},$$

where V_{DS} is the drain-source voltage, V_{GS} is the gate-source voltage and V_{TH} is the threshold voltage.

2. From the previous experiment, the 2N7000 transistor is presumed to be used in this experiment as well. Since the value $K = \frac{1}{2}\mu_n C_{ox} \frac{W}{L}$ is unknown, we can consider the value from LTspice.

$$K = 0.085$$

DC analysis of the circuit:

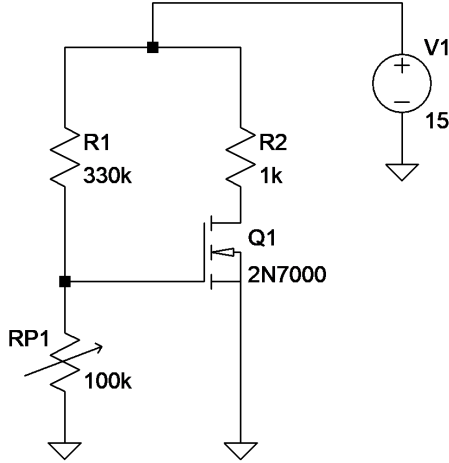


Figure 2: DC Analysis

To make sure $V_{GS} > V_{TH}$, inspect the $G - S$ loop.

$$V_G = V_1 \cdot \frac{R_{P1}}{R_1 + R_{P1}}, \quad V_S = 0 \text{ V}$$

$$\begin{aligned} V_{GS} > V_{TH} &\Rightarrow V_G - V_S = V_1 \cdot \frac{R_{P1}}{R_1 + R_{P1}} - 0 > V_{TH} \\ &\Rightarrow 15R_{P1} > 1.6R_{P1} + 1.6R_1 \Rightarrow 13.4R_{P1} > 1.6R_1 \\ &\Rightarrow R_{P1} > \frac{1.6}{13.4}R_1 \approx 39.4 \text{ k}\Omega \end{aligned}$$

Inspect the $D - S$ loop:

$$\begin{aligned} V_1 - I_D \cdot R_2 - V_{DS} &= 0, \quad V_{DS} \geq V_{GS} - V_{TH} \Rightarrow V_1 - I_D \cdot R_2 \geq V_{GS} - V_{TH} \\ \Rightarrow V_1 &\geq KR_2(V_{GS} - V_{TH})^2 + V_{GS} - V_{TH} \Rightarrow 85(V_{GS} - 1.6)^2 + V_{GS} - 1.6 \leq 15 \\ 85V_{GS}^2 - 271V_{GS} + 201 &\leq 0 \Rightarrow 1.174 \text{ V} \leq V_{GS} \leq 2.014 \text{ V} \\ \Rightarrow 1.174 &\leq V_1 \cdot \frac{R_{P1}}{R_1 + R_{P1}} \leq 2.014 \Rightarrow 28.2 \text{ k}\Omega \leq R_{P1} \leq 51.0 \text{ k}\Omega \end{aligned}$$

$$V_{GS} > V_{TH} \quad \& \quad V_{DS} \geq V_{GS} - V_{TH} \Rightarrow \boxed{39.4 \text{ k}\Omega \leq R_{P1} \leq 51.0 \text{ k}\Omega}$$

We want to achieve $V_{out} = 7.5 \text{ V}$.

$$\begin{aligned} V_{out} = V_{DS} &= V_1 - I_D \cdot R_2 = V_1 - KR_2 \left(V_1 \cdot \frac{R_{P1}}{R_1 + R_{P1}} - V_{TH} \right)^2 \\ &= 15 - 0.085 \cdot 1000 \left(15 \cdot \frac{R_{P1}}{330000 + R_{P1}} - 1.6 \right)^2 = 7.5 \\ \Rightarrow R_{P1,1} &\approx 31.34 \text{ k}\Omega, \quad R_{P1,2} \approx 47.74 \text{ k}\Omega \end{aligned}$$

Since 31.34 k Ω is out of range, $\boxed{R_{P1} = 47.74 \text{ k}\Omega}$.

Small-signal AC analysis of the circuit:

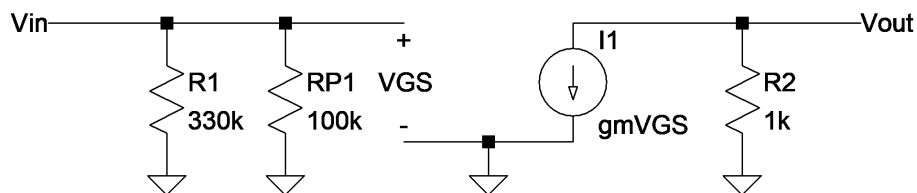


Figure 3: Small-Signal AC Analysis

$$A_v = \frac{V_{out}}{V_{in}} = \frac{-g_m V_{GS} R_2}{V_{GS}} = -2K(V_{GS} - V_{TH})R_2 = -2K \left(V_1 \cdot \frac{R_{P1}}{R_1 + R_{P1}} - V_{TH} \right) R_2$$

$$A_v = -2 \cdot 0.085 \left(15 \cdot \frac{47.74 \text{ k}}{330 \text{ k} + 47.74 \text{ k}} - 1.6 \right) \cdot 1000 \approx \boxed{-50.29}$$

Simulation

1. The circuit:

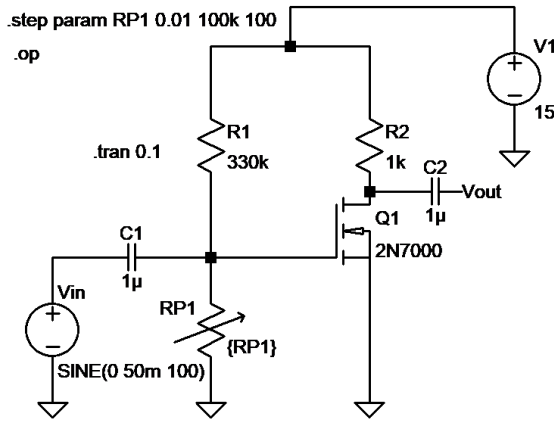


Figure 4: Circuit Diagram

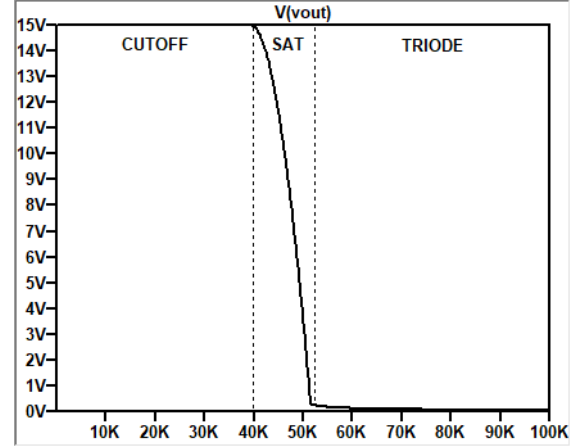


Figure 5: Graph for Different Values of R_{P1}

2. From Figure 5, we can observe that for $V_{out} = 7.5 \text{ V}$, the value of the resistor should be $R_{P1} \approx 47.94 \text{ k}\Omega$.

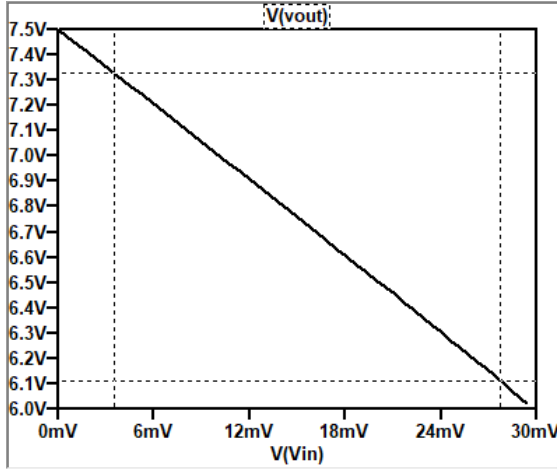


Figure 6: Circuit Behavior for 1 Millisecond

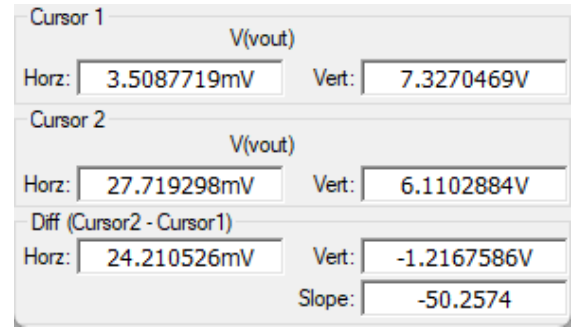


Figure 7: Gain (Slope) Obtained from Figure 6